

Arnaldo Silva

Electrical and Computer Engineer

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Summary

Dedicated and hard working individual driven to provide quality work and ethically inclined to find solutions that are both innovative and sustainable. As a team member, I strive to adapt to new and changing environments while collaborating with a variety of individuals to find the perfect solution. Having 4+ years experience in the utility space, I am interested in applying what I have learned in a fast-paced environment where I can continue to develop.

Education

Bachelor of Science in Electrical and Computer Engineering

Suffolk University at Boston, MA
9/2015 - 5/2019 3.2 / 4.0 GPA

Master of Science in Electrical Engineering

Clarkson University at Potsdam, NY
12/2019 - 6/2022 3.9 / 4.0 GPA
Advanced Certificate in Power Systems Engineering

Skills

C++
Intermediate

Swift
Intermediate

SQL
Proficient

Java
Proficient

Visual Basics
Intermediate

Machine Language / Assembly
Beginner

MATLab / Simulink
Intermediate

Multisim
Proficient

Microsoft Office Suite
Proficient

Python
Proficient

Professional Experience

Reliability Analytics Engineer

National Grid | September 2020 - Present

- Leading data driven discussions and monthly meetings on the company's key performance metrics related to reliability of our electrical grid.
- Maintaining databases used to contact our most critical facilities during outages.
- Automated workflows through the use of macros and visual basics to send weekly emails on the current performance of our at risk circuits.
- Automated the group's most used workbooks by integrating Oracle SQL based queries and macros, allowing managers to present year-to-date data.
- Provide visual interpretations of data that allow for quick understanding.
- Partnered with other functions of the company to plan and execute strategies to avoid state penalties on failing circuits.
- Maintain and update transmission and distribution circuit related databases.

Associate Reliability Analytics Engineer

National Grid | June 2019 - August 2020

- Entered the company through the Graduate Development Program (GDP) where hired individuals are trained on leadership skills and toured through various functions of the company.
- Supported project management of EV Central, a program designed to encourage employees to switch to an electric vehicle.
- Adopted responsibility over yearly regulatory filings involving reporting on utility reliability metrics such as SAIDI, SAIFI, and CAIDI. Reporting also included analysis on circuit characteristics for the Massachusetts division of the business.
- Created job continuity aids by documenting the department's responsibilities and instructions for each duty.

Intern

National Grid | May 2018 - December 2018

- Supported distributed energy resource projects by researching best practices for interconnecting renewables.
- Aided in starting an internal program that incentivized employees to switch to an electric vehicle and facilitated installation of charging stations at company owned buildings. I also provided data on the company's carbon reduction based on employee participation.
- Prepared the company on their pitch to get state regulatory approval on the Advanced Metering Infrastructure (AMI) project.
- Supported project management on preliminary studies on the cost - benefit of pitched ideas.

Related Experiences

Morse Decoder (03/2017 - 04/2017)

- Using a raspberry pi, coded a board to interpret morse code and output that code in a readable document.

Peer to Peer IP Communication (04/2018 - 05/2018)

- Coded a C++ program capable of connecting two computers using their IP addresses to send encrypted messages to each other and send any type of file

Smart Shoes (08/2018 - 05/2019)

- Developed software that tracked health metrics like heart rate, calories burned, and weight through hardware I embedded into running shoes using conductive thread.

Labs (08/2015 - 05/2019)

- Each of the ECE courses was paired with a lab where I utilized and became familiar with equipment such as multimeters, oscilloscopes, motors, batteries, and semiconductors.